Online Learning — Lecture Notes, 9 November 2026

Online gradient descent, quadratic regularization, and inventory decisions

1 Review: Online Gradient Descent

1.1 Forecasting a time series without observing the future

Consider a time series $(Y_t)_{t\geq 1}$ with observations $y_1, y_2, \ldots$. At round t, the objective is to predict y_t using only information available *before* observing it. Once y_t is revealed, the prediction rule can be updated.

In the simplest model, the learner selects $\theta_t \in \mathcal{K} \subseteq \mathbb{R}$ and predicts $\widehat{y}_t = \theta_t$. The feasible set $\mathcal{K}$ is assumed nonempty, closed, and convex.

Definition 1.1 (Instantaneous squared loss)**.** For an observation y_t, define

$$\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2, \qquad \ell_t'(\theta) = \theta - y_t.$$

The derivative is taken with respect to the *decision*. The realized loss $\ell_t(\theta_t)$ is only available after observing y_t.

extbfCaution. The *instantaneous* loss ℓ_t does not sum over rounds. Summing losses over $t = 1, \ldots, T$ defines the *cumulative* loss. Also note the sign: $\ell_t'(\theta_t) = \theta_t - y_t$, not $y_t - \theta_t$.

1.2 The OGD protocol: decision, loss, update

Online Gradient Descent (OGD) updates its decision after each observation. For nondifferentiable convex losses, gradients can be replaced by subgradients.

Definition 1.2 (Online Gradient Descent)**.** Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. Initialize $\theta_1 \in \mathcal{K}$ and choose step sizes $\eta_t > 0$. At round t:

1. **Decide:** select θ_t using past information only.
2. **Observe:** reveal y_t, compute $\ell_t(\theta_t)$, and select $g_t \in \partial \ell_t(\theta_t)$.
3. **Update:** set
$$\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t),$$
where $\Pi_{\mathcal{K}}$ denotes Euclidean projection onto $\mathcal{K}$.

One representation of the available history is $\mathcal{H}_{t-1} = \{(\theta_s, y_s, \ell_s(\theta_s), g_s) : s < t\}$. The decision θ_t may depend on this history, but not on y_t.

For scalar squared loss,

$$\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t(\theta_t - y_t)) = \Pi_{\mathcal{K}}((1-\eta_t)\theta_t + \eta_t y_t).$$

If $\mathcal{K} = \mathbb{R}$ and $0 < \eta_t \leq 1$, the update is a weighted average of the previous prediction and the latest observation. It may be used to predict y_{t+1}, but was not used to predict y_t.

2 OGD for Regression with Covariates

2.1 A sequentially updated linear predictor

Consider a bakery forecasting daily baguette sales. Past sales and a promotion announced in advance may be available before the day's demand is observed.

Let $z_t \in \mathbb{R}^d$ be the feature vector available at the start of day t, and let $w_t \in \mathcal{K} \subseteq \mathbb{R}^d$ denote the

model coefficients. Define

$$\widehat{y}_t = w_t^\top z_t, \qquad \ell_t(w) = \frac{1}{2}(w^\top z_t - y_t)^2.$$

The model and loss function determine the gradient used by OGD.

Proposition 2.1 (Gradient of the regression loss). *For* $\ell_t(w) = \frac{1}{2}(w^\top z_t - y_t)^2$,

$$g_t = \nabla \ell_t(w_t) = \underbrace{(w_t^\top z_t - y_t)}_{e_t} z_t, \qquad w_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta_t e_t z_t),$$

where $e_t = \widehat{y}_t - y_t$ *is the signed prediction error.*

Proof. Apply the chain rule and use $\nabla_w(w^\top z_t) = z_t$. □

Remark 2.2 (One-step-ahead forecasting chronology). At the end of day t, the observations $y_1, \dots, y_t$ are known. The next forecast uses the updated coefficients and features available before the next target is observed:

$$\widehat{y}_{t+1|t} = w_{t+1}^\top z_{t+1}.$$

In particular, z_{t+1} must not contain the unknown target y_{t+1}.

2.2 Temporal lags and harmonic features

A fixed-dimensional feature vector may contain the previous p sales and a promotion indicator $x_t \in \{0, 1\}$:

$$z_t^{\text{lags}} = (1, x_t, y_{t-1}, \dots, y_{t-p})^\top, \qquad t \geq p + 1.$$

The leading 1 represents the intercept. Unlike the full history $(y_1, \dots, y_{t-1})$, the feature dimension remains fixed.

To summarize oscillations in the historical window, one may compute Fourier coefficients. For $p \geq 7$ and $k \in \{1, 2, 3\}$, one convention is

$$a_{k,t} = \frac{2}{p} \sum_{j=1}^{p} y_{t-j} \cos\left(\frac{2\pi k(j-1)}{p}\right),$$
$$b_{k,t} = \frac{2}{p} \sum_{j=1}^{p} y_{t-j} \sin\left(\frac{2\pi k(j-1)}{p}\right).$$

These coefficients are available before observing y_t. One could define

$$z_t = (1, x_t, y_{t-1}, \dots, y_{t-p}, a_{1,t}, b_{1,t}, a_{2,t}, b_{2,t}, a_{3,t}, b_{3,t})^\top.$$

extbfCaution. The Fourier coefficients above are **linear combinations of the same p lags**. Including them alongside *all* the original lags adds no information to a linear predictor and creates linearly dependent regression columns. Instead, one may replace lags with harmonic summaries, or add distinct calendar-based seasonal features.

For a known period P, calendar harmonics

$$c_{k,t} = \cos\left(\frac{2\pi kt}{P}\right), \qquad s_{k,t} = \sin\left(\frac{2\pi kt}{P}\right)$$

are known in advance and can complement lag features. Any feature normalization must also be fitted using past observations only.

2.3 Review: positive matrices and convexity

Definition 2.3 (Positive semidefinite and positive definite matrices)**.** Let $A \in \mathbb{R}^{d\times d}$ be **symmetric**.

- A is **positive semidefinite**, written $A \succeq 0$, if $v^\top Av \geq 0$ for every $v \in \mathbb{R}^d$.
- A is **positive definite**, written $A \succ 0$, if $v^\top Av > 0$ for every nonzero $v \in \mathbb{R}^d$.

By the spectral theorem, $A = Q\Lambda Q^\top$ for an orthogonal matrix Q and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$. Thus

$$A \succeq 0 \iff \lambda_i \geq 0 \ \forall i, \qquad A \succ 0 \iff \lambda_i > 0 \ \forall i.$$

Property 2.4 (Hessian criteria)**.** *Let $f : \Omega \to \mathbb{R}$ be twice continuously differentiable on an* open convex *set $\Omega \subseteq \mathbb{R}^d$. Then*

$$f \text{ is convex} \iff \nabla^2 f(w) \succeq 0 \quad \forall w \in \Omega.$$

If $\nabla^2 f(w) \succ 0$ everywhere, then f is strictly convex; this is sufficient but not necessary. Moreover, $\nabla^2 f(w) \succeq \alpha I_d$ everywhere characterizes α-strong convexity for $\alpha > 0$.

Example 2.5 (A rank-one Hessian)**.** For linear regression with squared loss,

$$\nabla^2 \ell_t(w) = z_t z_t^\top \succeq 0.$$

Its rank is at most one. For $d > 1$, it has zero eigenvalues and therefore is not strongly convex on $\mathbb{R}^d$, even though it is convex.

3 Regret of OGD Regression

3.1 Boundedness assumptions and a $\sqrt{T}$ bound

Assume

$$\mathcal{K} = \{w \in \mathbb{R}^d : \|w\|_2 \leq B\}, \quad \|z_t\|_2 \leq R, \quad |y_t| \leq Y,$$

with $B, R > 0$ and $Y \geq 0$. The diameter of $\mathcal{K}$ is $D = 2B$. For $w \in \mathcal{K}$, Cauchy–Schwarz yields

$$|w^\top z_t - y_t| \leq BR + Y, \qquad \|\nabla \ell_t(w)\|_2 \leq \underbrace{(BR+Y)R}_{G_{\text{quad}}}.$$

Hence the gradient bound requires bounded coefficients, features, and targets.

Theorem 3.1 (Square-root regret for projected OGD)**.** *Assume $G_{\text{quad}} > 0$ and a known horizon T. Initialize $w_1 \in \mathcal{K}$ and choose the constant step size*

$$\eta_t = \eta = \frac{2B}{G_{\text{quad}}\sqrt{T}}.$$

Then, for every fixed comparator $u \in \mathcal{K}$,

$$\operatorname{Reg}_T(u) \leq 2BG_{\text{quad}}\sqrt{T} = 2B(BR+Y)R\sqrt{T}.$$

The same upper bound holds for standard regret against the best fixed comparator.

Proof. By convexity and nonexpansiveness of Euclidean projection,

$$\ell_t(w_t) - \ell_t(u) \le \langle g_t, w_t - u \rangle \le \frac{\|w_t - u\|_2^2 - \|w_{t+1} - u\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

Summing and using $\|w_1 - u\|_2 \le D = 2B$ gives

$$\mathrm{Reg}_T(u) \le \frac{D^2}{2\eta} + \frac{\eta G_{\mathrm{quad}}^2 T}{2}.$$

Choosing $\eta = D/(G_{\mathrm{quad}}\sqrt{T})$ yields $\mathrm{Reg}_T(u) \le D G_{\mathrm{quad}}\sqrt{T}$. □

Dividing by T gives

$$\frac{\mathrm{Reg}_T}{T} \le \frac{2B(BR+Y)R}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0.$$

This establishes $\limsup_{T\to\infty} \mathrm{Reg}_T / T \le 0$: there is no asymptotically positive average excess loss relative to the best fixed comparator. Since an adaptive algorithm can achieve negative regret, a one-sided bound does not imply two-sided convergence of realized regret to zero.

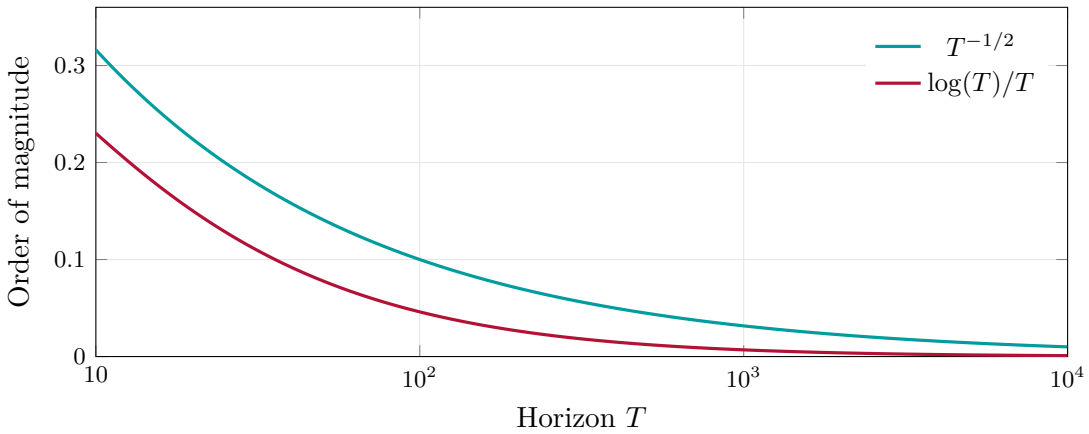
3.2 Comparing rates of average regret

Ignoring multiplicative constants, consider

$$T^{-1/2} \qquad \text{and} \qquad \frac{\log T}{T}.$$

The latter decreases faster, since

$$\frac{(\log T)/T}{T^{-1/2}} = \frac{\log T}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0.$$



Logarithmic horizontal axis; log denotes the natural logarithm.

T	$T^{-1/2}$	$(\log T)/T$
10^2	0.1000	0.04605
10^3	0.0316	0.00691
10^4	0.0100	0.00092

Remark 3.2 (Interpretation). An $O(\log T)$ regret bound implies an average bound of $O((\log T)/T)$, faster than $O(T^{-1/2})$. Such a result typically requires additional curvature assumptions, notably strong convexity. Quadratic regularization provides one way to obtain this property.

4 OGD with Quadratic Regularization

4.1 Adding a penalty to each loss

The squared regression loss is convex but, for $d > 1$, is not strongly convex at each round. Add a quadratic penalty on the model coefficients.

Definition 4.1 (Regularized squared loss)**.** For $\lambda > 0$, define

$$\widetilde{\ell}_t(w) = \underbrace{\frac{1}{2}(w^\top z_t - y_t)^2}_{\text{prediction error}} + \underbrace{\frac{\lambda}{2}\|w\|_2^2}_{\text{quadratic penalty}} .$$

The penalty is included *at every round*.

Its gradient and Hessian are

$$\nabla\widetilde{\ell}_t(w) = (w^\top z_t - y_t)z_t + \lambda w,$$
$$\nabla^2\widetilde{\ell}_t(w) = z_t z_t^\top + \lambda I_d \succeq \lambda I_d.$$

Proposition 4.2 (Strong convexity and regularized update)**.** *Each $\widetilde{\ell}_t$ is λ-strongly convex. Writing $e_t = w_t^\top z_t - y_t$, projected OGD becomes*

$$w_{t+1} = \Pi_{\mathcal{K}}\left(\underbrace{(1-\eta_t\lambda)w_t}_{\textit{coefficient shrinkage}} - \underbrace{\eta_t e_t z_t}_{\textit{data-driven correction}}\right).$$

The penalty pulls coefficients toward zero, whereas the data term corrects the current prediction error. Projection keeps parameters feasible.

extbfCaution. The penalty here applies to *all* coefficients, including any intercept, ensuring $\nabla^2\widetilde{\ell}_t \succeq \lambda I_d$ in every direction. Excluding the intercept, as is common in ridge regression, requires a separate strong-convexity analysis.

4.2 A logarithmic guarantee for regularized OGD

Keep the assumptions

$$\mathcal{K} = \{w : \|w\|_2 \le B\}, \quad \|z_t\|_2 \le R, \quad |y_t| \le Y, \quad \lambda > 0.$$

Then, for all $w \in \mathcal{K}$,

$$\|\nabla\widetilde{\ell}_t(w)\|_2 \le \underbrace{(BR+Y)R + \lambda B}_{G_\lambda}.$$

Theorem 4.3 (Logarithmic regret for regularized OGD)**.** *Initialize $w_1 \in \mathcal{K}$ and choose $\eta_t = 1/(\lambda t)$. For any fixed $u \in \mathcal{K}$,*

$$\sum_{t=1}^{T}[\widetilde{\ell}_t(w_t) - \widetilde{\ell}_t(u)] \le \frac{G_\lambda^2}{2\lambda}(1+\log T).$$

Thus standard regret for the regularized losses *satisfies*

$$\widetilde{\mathrm{Reg}}_T \le \frac{G_\lambda^2}{2\lambda}(1+\log T).$$

Proof. Strong convexity and the projection inequality yield

$$\begin{aligned}\widetilde{\ell}_t(w_t) - \widetilde{\ell}_t(u) &\le \langle g_t, w_t - u\rangle - \frac{\lambda}{2}\|w_t - u\|_2^2 \\ &\le \frac{\|w_t - u\|_2^2 - \|w_{t+1} - u\|_2^2}{2\eta_t} + \frac{\eta_t G_\lambda^2}{2} - \frac{\lambda}{2}\|w_t - u\|_2^2.\end{aligned}$$

With $\eta_t^{-1} = \lambda t$, the comparator-distance terms cancel when summed, apart from a nonpositive final term. Therefore

$$\sum_{t=1}^{T}[\widetilde{\ell}_t(w_t) - \widetilde{\ell}_t(u)] \le \frac{G_\lambda^2}{2\lambda}\sum_{t=1}^{T}\frac{1}{t} \le \frac{G_\lambda^2}{2\lambda}(1 + \log T).$$

□

Remark 4.4 (Which regret is bounded?). The theorem compares *regularized* losses $\widetilde{\ell}_t = \ell_t + \lambda\|w\|_2^2/2$. It does not automatically provide logarithmic regret for the unregularized squared losses ℓ_t. Moreover, adding the penalty at every round differs from FTRL with a single regularizer added to the cumulative historical loss.

5 Inventory Management

5.1 Motivation: excess inventory and shortages

Inventory management requires choosing an order or production quantity before actual demand is known. At round t, choose $q_t \in \mathcal{K} = [0, Q_{\max}]$ and then observe demand $y_t \ge 0$.

Two economic costs arise:

- **Excess inventory:** $c_h > 0$ is the unit *holding cost*, which may include storage, unsold goods, spoilage, and markdowns.
- **Shortage:** $c_b > 0$ is the unit *shortage cost*, which may represent lost margins, service penalties, or backorder costs.

Write the decision error as $\delta_t = q_t - y_t$. With $(a)_+ = \max\{a, 0\}$, the cost components are

$$\underbrace{c_h(q_t - y_t)_+}_{\text{excess inventory}}, \qquad \underbrace{c_b(y_t - q_t)_+}_{\text{shortage}}.$$

Definition 5.1 (Inventory loss: newsvendor)**.** The inventory loss at round t is

$$\ell_t^{\text{inv}}(q) = c_h(q - y_t)_+ + c_b(y_t - q)_+. \tag{1}$$

This function is **convex**, but is not differentiable at $q = y_t$.

Remark 5.2. The *newsvendor* model historically refers to newspaper sales: orders must be placed before demand is known, while unsold copies quickly lose value. We consider a single-period model with no inventory carryover and no replenishment lead time.

5.2 From absolute loss to pinball loss

Recall the absolute or L^1 loss:

$$\ell_1(\widehat{y}, y) = |\widehat{y} - y|.$$

For a parametric model μ_β, empirical absolute-risk minimization is

$$\widehat{\beta} \in \arg\min_{\beta\in\mathbb{R}^p} \sum_{i=1}^{n} |y_i - \mu_\beta(x_i)|.$$

For a constant predictor, the minimizers are the empirical medians:

$$\arg\min_{c\in\mathbb{R}} \sum_{i=1}^{n} |y_i - c| = \mathrm{Med}(y_1, \dots, y_n).$$

Remark 5.3. When n is even, the minimizer need not be unique: any point between the two central order statistics is optimal. In general regression, absolute-loss minimization targets a **conditional median**, not the unconditional median of all observations.

Definition 5.4 (Pinball loss)**.** For $\tau \in (0,1)$, define

$$\rho_\tau(q, y) = (1-\tau)(q-y)_+ + \tau(y-q)_+. \tag{2}$$

The parameter τ controls the asymmetry between overprediction and underprediction costs.

At $\tau = 1/2$, the absolute loss is recovered up to a factor:

$$\rho_{1/2}(q, y) = \frac{1}{2}|q-y|.$$

Thus pinball loss generalizes absolute loss to asymmetric penalties.

5.3 The newsvendor–pinball identity

Starting from inventory loss,

$$\begin{aligned}\ell_t^{\mathrm{inv}}(q) &= c_h(q-y_t)_+ + c_b(y_t-q)_+ \\ &= (c_h+c_b)\left[\frac{c_h}{c_h+c_b}(q-y_t)_+ + \frac{c_b}{c_h+c_b}(y_t-q)_+\right].\end{aligned}$$

Set

$$\tau = \frac{c_b}{c_h+c_b}, \qquad 1-\tau = \frac{c_h}{c_h+c_b}.$$

Proposition 5.5 (Newsvendor–pinball identity)**.** *For any q and demand y_t,*

$$\ell_t^{\mathrm{inv}}(q) = (c_h+c_b)\rho_\tau(q, y_t), \qquad \tau = \frac{c_b}{c_h+c_b}. \tag{3}$$

Example 5.6 (A shortage is three times as costly)**.** Suppose $c_b = 3c_h$. Then

$$\tau = \frac{3c_h}{3c_h + c_h} = \frac{3}{4} = 0.75.$$

One unit of excess inventory costs c_h, whereas one unit of unmet demand costs $3c_h$. Under the probabilistic model introduced below, the optimal decision is the **75th percentile** of demand.

5.4 Optimal decision: the critical quantile

Let demand Y be an integrable random variable with cumulative distribution function

$$F_Y(q) = \mathbb{P}(Y \le q), \qquad F_Y(q^-) = \mathbb{P}(Y < q).$$

We aim to minimize expected cost:

$$\mathcal{C}(q) = \mathbb{E}\left[c_h(q-Y)_+ + c_b(Y-q)_+\right].$$

Proposition 5.7 (Optimal newsvendor quantile)**.** *Let $c_h, c_b > 0$ and $\tau = c_b/(c_h + c_b)$. A quantity $q^\star$ minimizes expected cost over $\mathbb{R}$ if and only if*

$$F_Y(q^{\star-}) \le \tau \le F_Y(q^\star). \tag{4}$$

In particular, one optimal choice is the generalized τ-quantile:

$$q^\star = F_Y^{-1}(\tau) := \inf\{q \in \mathbb{R} : F_Y(q) \ge \tau\}.$$

If F_Y is continuous and strictly increasing near this quantile, the optimum is unique.

Proof. The function $\mathcal{C}$ is convex. Its left and right derivatives are

$$\mathcal{C}'_-(q) = (c_h + c_b)F_Y(q^-) - c_b, \qquad \mathcal{C}'_+(q) = (c_h + c_b)F_Y(q) - c_b.$$

The convex optimality condition is $\mathcal{C}'_-(q^\star) \le 0 \le \mathcal{C}'_+(q^\star)$, which is equivalent to

$$F_Y(q^{\star-}) \le \frac{c_b}{c_h + c_b} \le F_Y(q^\star).$$

□

Remark 5.8 (Capacity constraint). If $Y \ge 0$ and $\mathcal{K} = [0, Q_{\max}]$, one may choose

$$q^\star_{\mathcal{K}} = \min\{Q_{\max}, F_Y^{-1}(\tau)\}.$$

The capacity constraint may prevent ordering the unconstrained critical quantile.

5.5 Application to Online Learning

In online learning, the distribution of Y need not be known, and demands $y_1, \dots, y_T$ are revealed sequentially. The learner selects q_t **before** observing y_t and incurs $\ell_t^{\text{inv}}(q_t)$.

Regret against the best fixed quantity in hindsight is

$$\text{Reg}_T^{\text{inv}} = \sum_{t=1}^T \ell_t^{\text{inv}}(q_t) - \min_{q \in \mathcal{K}} \sum_{t=1}^T \ell_t^{\text{inv}}(q).$$

Property 5.9 (Subgradient of inventory loss)**.** *The subdifferential is*

$$\partial \ell_t^{\text{inv}}(q) = \begin{cases} \{-c_b\}, & q < y_t, \\ [-c_b, c_h], & q = y_t, \\ \{c_h\}, & q > y_t. \end{cases}$$

Every subgradient satisfies

$$|g_t| \le G := \max\{c_h, c_b\}.$$

Proof. Use $(a)_+ = (a + |a|)/2$ to obtain

$$\ell_t^{\text{inv}}(q) = \frac{c_h + c_b}{2}|q - y_t| + \frac{c_h - c_b}{2}(q - y_t).$$

By subdifferential calculus,

$$\partial \ell_t^{\text{inv}}(q) = \frac{c_h + c_b}{2}\partial|q - y_t| + \frac{c_h - c_b}{2}.$$

Using $\partial|z| = \{-1\}$ for $z < 0$, $[-1, 1]$ for $z = 0$, and $\{1\}$ for $z > 0$ gives the three cases. Finally, $\partial \ell_t^{\text{inv}}(q) \subseteq [-c_b, c_h]$. □

Projected subgradient descent is therefore applicable:

$$q_{t+1} = \Pi_{\mathcal{K}}(q_t - \eta_t g_t), \qquad g_t \in \partial \ell_t^{\mathrm{inv}}(q_t).$$

If $q_t < y_t$, then $g_t = -c_b$ and the order increases. If $q_t > y_t$, then $g_t = c_h$ and the order decreases.

Remark 5.10 (FTL and the empirical quantile). For inventory loss, FTL minimizes cumulative past loss:

$$q_t \in \arg\min_{q \in \mathcal{K}} \sum_{s=1}^{t-1} \ell_s^{\mathrm{inv}}(q).$$

When the capacity constraint is inactive, one may select the empirical τ-quantile of $y_1, \dots, y_{t-1}$. Thus FTL with squared loss produces a historical mean, whereas FTL with pinball loss produces a historical quantile.

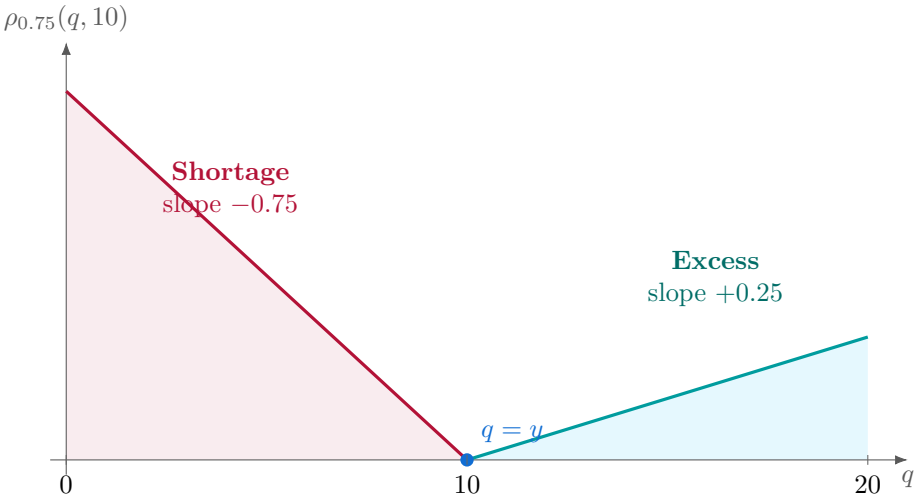
5.6 Geometric representation of pinball loss

Consider $\tau = 0.75$:

$$\rho_{0.75}(q, y) = 0.25(q - y)_+ + 0.75(y - q)_+.$$

Fix $y = 10$. Then

$$\rho_{0.75}(q, 10) = \begin{cases} 0.75(10 - q), & q < 10, \\ 0, & q = 10, \\ 0.25(q - 10), & q > 10. \end{cases}$$



Pinball loss for $\tau = 0.75$ and $y = 10$. Underprediction is penalized three times as much as overprediction.

Remark 5.11 (Economic interpretation). A unit of unmet demand has normalized cost 0.75, compared with 0.25 for a unit of excess inventory. Their ratio is

$$\frac{c_b}{c_h} = \frac{0.75}{0.25} = 3.$$

Thus τ directly controls loss asymmetry.

5.7 OGD: an operational correction of orders

Initialize $q_1 \in [0, Q_{\max}]$. After observing y_t, select $g_t \in \partial \ell_t^{\mathrm{inv}}(q_t)$. The correction depends on the observed outcome:

Outcome	$\partial \ell_t(q_t)$	g_t	Next order before projection
Shortage: $q_t < y_t$	$\{-c_b\}$	$-c_b$	$q_t + \eta_t c_b$
Exact match: $q_t = y_t$	$[-c_b, c_h]$	0	q_t
Excess: $q_t > y_t$	$\{c_h\}$	c_h	$q_t - \eta_t c_h$

Definition 5.12 (Projected OGD update)**.** The order update is

$$q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t - \eta_t g_t), \tag{5}$$

where $\eta_t > 0$ is the learning rate and

$$\Pi_{[0,Q_{\max}]}(z) = \min\{Q_{\max}, \max\{0, z\}\}.$$

Remark 5.13 (Operational interpretation). A **shortage** raises the next order; **excess inventory** lowers it. Relative costs c_b and c_h determine the sizes of these corrections, while projection enforces capacity constraints. At $q_t = y_t$, we choose the valid subgradient $g_t = 0$. Before projection, the correction magnitude depends on the unit cost and step size, **not on the error magnitude** $|q_t - y_t|$.

5.8 OGD regret for inventory decisions

Consider the bounded convex set

$$\mathcal{K} = [0, Q_{\max}], \qquad Q_{\max} > 0,$$

whose Euclidean diameter is $D = Q_{\max}$. All inventory subgradients satisfy $|g_t| \le G := \max\{c_b, c_h\}$.

Regret against the best **fixed quantity in hindsight** is

$$\mathrm{Reg}_T^{\mathrm{inv}} = \sum_{t=1}^{T} \ell_t^{\mathrm{inv}}(q_t) - \min_{q \in \mathcal{K}} \sum_{t=1}^{T} \ell_t^{\mathrm{inv}}(q).$$

Theorem 5.14 (OGD regret for inventory management)**.** *Assume convex losses, $\mathcal{K} = [0, Q_{\max}]$, and $|g_t| \le G$. For a constant step size $\eta > 0$, projected OGD satisfies*

$$\mathrm{Reg}_T^{\mathrm{inv}} \le \frac{Q_{\max}^2}{2\eta} + \frac{\eta G^2 T}{2}. \tag{6}$$

For a known horizon T, choose

$$\eta = \frac{Q_{\max}}{G\sqrt{T}}, \qquad \mathrm{Reg}_T^{\mathrm{inv}} \le Q_{\max} G \sqrt{T}.$$

In particular,

$$\frac{\mathrm{Reg}_T^{\mathrm{inv}}}{T} \le \frac{Q_{\max} G}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0, \qquad \limsup_{T\to\infty} \frac{\mathrm{Reg}_T^{\mathrm{inv}}}{T} \le 0.$$

Proof. By nonexpansiveness of Euclidean projection, for any $u \in \mathcal{K}$,

$$|q_{t+1} - u|^2 \le |q_t - \eta g_t - u|^2 = |q_t - u|^2 - 2\eta g_t(q_t - u) + \eta^2 g_t^2.$$

Convexity gives $\ell_t^{\mathrm{inv}}(q_t) - \ell_t^{\mathrm{inv}}(u) \le g_t(q_t - u)$, hence

$$\ell_t^{\mathrm{inv}}(q_t) - \ell_t^{\mathrm{inv}}(u) \le \frac{|q_t - u|^2 - |q_{t+1} - u|^2}{2\eta} + \frac{\eta}{2} g_t^2.$$

Summing over $t = 1, \dots, T$ telescopes the squared distances:

$$\mathrm{Reg}_T^{\mathrm{inv}}(u) \leq \frac{|q_1 - u|^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} g_t^2 \leq \frac{Q_{\max}^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

The inequality holds for every fixed $u \in \mathcal{K}$, including the best fixed comparator in hindsight. □

Remark 5.15 (Interpretation of the guarantee). The comparison is between:

- **Online decisions:** $q_1, \dots, q_T$, chosen before the corresponding demands are observed.
- **Hindsight benchmark:** a single feasible quantity $q^\star$ selected after observing the entire sequence.

The comparator is a feasible empirical τ-quantile, with $\tau = c_b/(c_h + c_b)$, clipped to $[0, Q_{\max}]$ if necessary. The bound requires no stationarity, independence, or known demand distribution. It controls asymptotically positive average excess cost, but does not compare against a sequence of changing conditional quantiles.

Key takeaway

Inventory management and online learning. Pinball loss incorporates asymmetric economic costs into decisions. OGD adjusts orders after observed shortages and excesses while respecting operational constraints. On a bounded feasible set, convexity and bounded subgradients give the one-sided regret guarantee

$$\mathrm{Reg}_T^{\mathrm{inv}} \leq O(\sqrt{T}).$$

Cumulative performance is controlled against the best fixed quantity, without assuming a probabilistic model for demand.

6 General Conclusion

Online learning chooses a decision before the current observation is revealed, then updates that decision from the observed loss. The methods studied differ in their update rules:

- **FTL:** minimizes cumulative past losses.
- **FTRL:** adds regularization to stabilize decisions.
- **OGD:** makes a correction using the current (sub)gradient.

The loss function encodes the statistical target:

$$\begin{aligned} \text{Squared loss} &\longrightarrow \text{mean}, \\ \text{Absolute loss} &\longrightarrow \text{median}, \\ \text{Pinball loss} &\longrightarrow \text{quantile}. \end{aligned}$$

These correspondences apply to constant prediction, or to their conditional analogues under appropriate assumptions.

In inventory management, the choice of loss is not merely statistical: **it directly encodes the economic costs of excess stock and shortages**.

$$\tau = \frac{c_b}{c_h + c_b} \quad \Longrightarrow \quad q^\star = F_Y^{-1}(\tau).$$

Finally, distinguish the **probabilistic framework**, which minimizes expected cost, from the **online framework**, which controls cumulative regret without requiring a known demand distribution.